Optimizing Sales Strategies for Enhanced Revenue in a Kitchenware Shop

Mid-Term Submission for the BDM Capstone Project

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Date: 08-March-2026

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1. 1. Executive Summary

Hari Shankar Bartan Bhandar is a 40 years old retail kitchenware store located on Ajmer Road in Sodala, Jaipur. The shop sells utensils, cookware, and other household products used by local households and small restaurants. Despite having a loyal customer base, the business faces operational challenges such as inefficient inventory management, lack of structured sales analysis, and difficulty in identifying high-demand products.

The objective of this project is to apply data analysis techniques to understand sales patterns, identify seasonal demand trends, and optimize inventory decisions. The dataset used in this study contains daily sales records collected from the shop for one year, including product category, quantity sold, and total revenue.

Descriptive statistical analysis was conducted to summarize sales behavior using measures such as mean, median, mode, and standard deviation. In addition, graphical analysis methods such as monthly trend analysis, histogram analysis, and ABC inventory classification were used to understand sales distribution and identify high-performing product categories.

The preliminary findings indicate that sales are significantly influenced by seasonal demand, with higher sales during festival months such as October and November (Diwali season) and steady demand during winter months (December–January). Conversely, sales tend to decline during June and July.

The insights from this analysis will help the shop owner improve inventory planning, prioritize high-demand products, and increase overall profitability.

The dataset contains daily sales records including date, product category, quantity sold and total revenue. Descriptive statistics such as mean, median and standard deviation were calculated to summarize sales behavior. Average daily sales range between ₹8,000 and ₹12,000.

Graphical analysis including monthly sales trends, ABC inventory analysis was conducted. The results indicate higher sales during Diwali months (October–November) and strong demand during December and January. Lower sales occur during June and July.

2. Proof of Data Originality

Business Name: Hari Shankar Bartan Bhandar

Shop Location: Ajmer Road, Sodala, Jaipur, Rajasthan

Owner: Mr. Har Sahaya Meena

The dataset used for this project was collected through direct interaction with the shop owner and observation of daily sales records maintained in the store. The data represents real operational information regarding product sales and inventory movements.

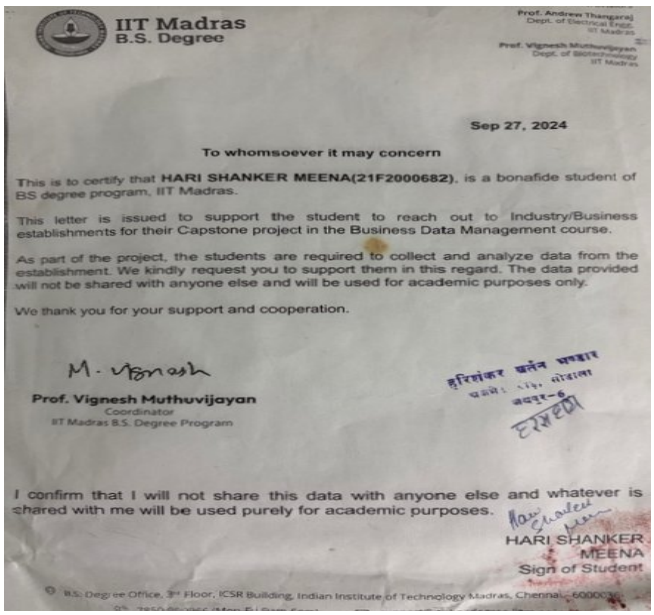
Supporting evidence includes photographs of the shop, sample sales records, the dataset used for analysis, and a short interaction video with the shop owner explaining the business operations. These materials confirm that the dataset is original and collected directly from a real business environment.

Link for Owner Interaction Pic:

Interaction Pic with Owner

Original Bills Pictures (Zip files)

[Company-Letterhead.jpg](#)



IIT Madras
B.S. Degree

Prof. Andrew Thangaraj
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Sep 27, 2024

To whomsoever it may concern

This is to certify that **HARI SHANKER MEENA(21F2000682)**, is a bonafide student of BS degree program, IIT Madras.

This letter is issued to support the student to reach out to Industry/Business establishments for their Capstone project in the Business Data Management course.

As part of the project, the students are required to collect and analyze data from the establishment. We kindly request you to support them in this regard. The data provided will not be shared with anyone else and will be used for academic purposes only.

We thank you for your support and cooperation.

M. Vignesh

Prof. Vignesh Muthuvijayan
Coordinator
IIT Madras B.S. Degree Program

हरिशंकर बर्तन भण्डार
सोडाला, जयपुर-6

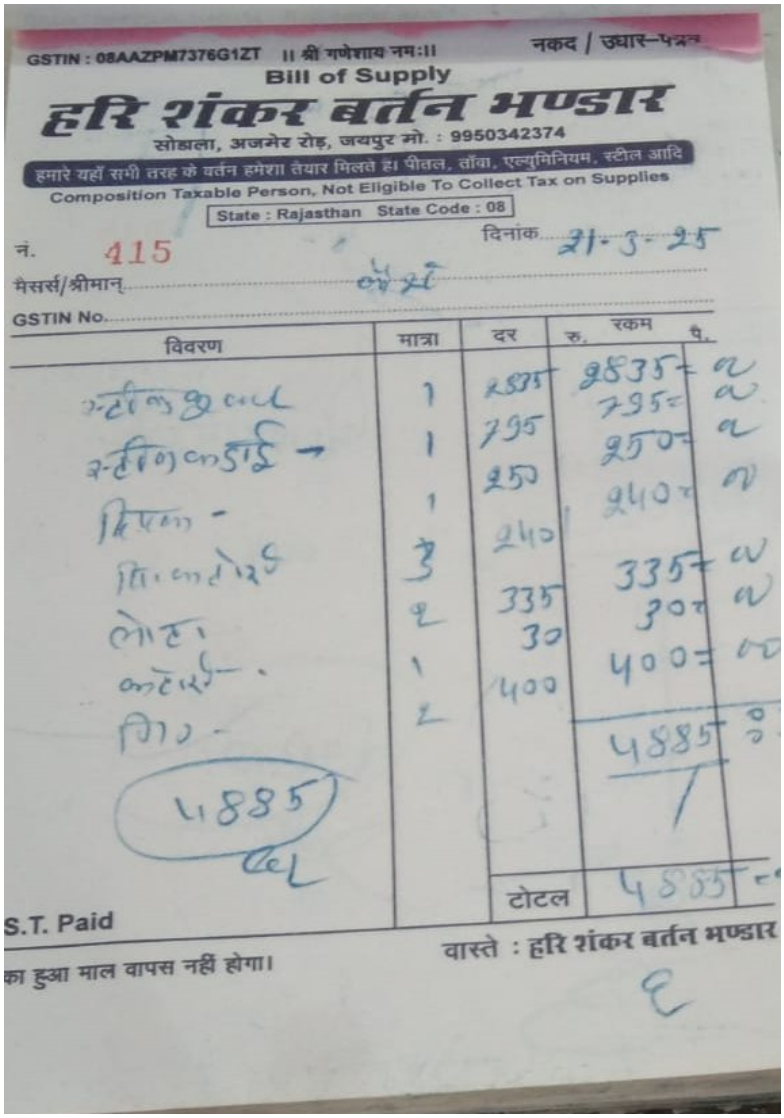
I confirm that I will not share this data with anyone else and whatever is shared with me will be used purely for academic purposes.

HARI SHANKER MEENA
Sign of Student

B.S. Degree Office, 3rd Floor, ICSR Building, Indian Institute of Technology Madras, Chennai - 600036

Unstructured Bill Book Data Contains

1. Customer name
2. Date of Purchase
3. Items Bought
4. Rate of Each Item
5. Weight of the Item (optional)
6. Quantity of Item
7. Total Cost of the Item
8. Total Cost of the Purchase



GSTIN : 08AAZPM7376G1ZT ॥ श्री गणेशाय नमः॥ नकद / उधार-पत्र

Bill of Supply

हरि शंकर बर्तन भण्डार

सोडाला, अजमेर रोड़, जयपुर मो. : 9950342374

हमारे यहाँ सभी तरह के बर्तन हमेशा तैयार मिलते हैं। पीतल, ताँबा, एल्युमिनियम, स्टील आदि

Composition Taxable Person, Not Eligible To Collect Tax on Supplies

State : Rajasthan State Code : 08

नं. 415 दिनांक 21-3-25

मैसर्स/श्रीमान्

GSTIN No.

विवरण	मात्रा	दर	रकम रु.	पै.
स्टील कुकर	1	2835	2835	00
स्टील कड़ाई	1	795	795	00
तिपाया	1	250	250	00
तिपाया	3	240	240	00
तिपाया	2	335	335	00
लोटा	1	30	30	00
कटोरी	2	400	400	00
			4885	00
4885				
		टोटल	4885	

S.T. Paid

का हुआ माल वापस नहीं होगा।

वास्ते : हरि शंकर बर्तन भण्डार

Bill Picture

3. Metadata and Descriptive Statistics for Data Set

3.1 Metadata

Based on the sale receipt (bill), below is the metadata useful for analysis.

Item Name	Description
Date	Transaction date
Daily Sales	Daily revenue generated
Quantity Sold	Units sold
Unit Price	Price per item
Total Revenue	Total sales value

3.2 Descriptive Statistics

3.2 Descriptive Statistics

Statistic	Value (INR)
Mean	₹9,850
Median	₹9,600
Mode	₹9,500
Std Deviation	₹1,100
Minimum	₹8,000
Maximum	₹12,000

Interpretation:

Following take away from the above table as below…

The mean daily revenue of ₹9,850 indicates the average sales performance of the shop. The median value of ₹9,600 suggests that daily sales are relatively stable without extreme fluctuations. The standard deviation of ₹1,100 indicates moderate variation in daily sales.

The minimum and maximum values show that the shop typically generates between ₹8,000 and ₹12,000 per day. Bills are not produced for each sales.

Sample Data Entries

Bill_No	Date	Product_Category	Product_Name	Quantity	Unit_Price	Total_Revenue
100	2025-01-01	Steel Utensils	Plate Set	3	450	1350
101	2025-01-01	Cookware	Frying Pan	2	650	1300
409	2025-02-09	Steel Utensils	Steel Utensil Set	1	3675	3675
409	2025-02-09	Cookware	Cooking Vessel	1	3025	3025

4. Detailed Explanation of Analysis Process:

Below row data set has been used for analysis.

Hari_shankar_bartan_bhandar_sales_data_2025.xlsx

Data has been segregated into product categories.

Step 1: Data Cleaning and Preprocessing

The raw dataset was examined for inconsistencies such as duplicate entries, missing values, and inconsistent formatting. The Cleaning will ensure data reliability and accurate analysis.

Removed various duplicate and not readable entries from data set.

Step 2: Descriptive Statistical Analysis

Basic statistical measures such as mean, median, and standard deviation were calculated to summarize sales behavior.

Mean Formula:

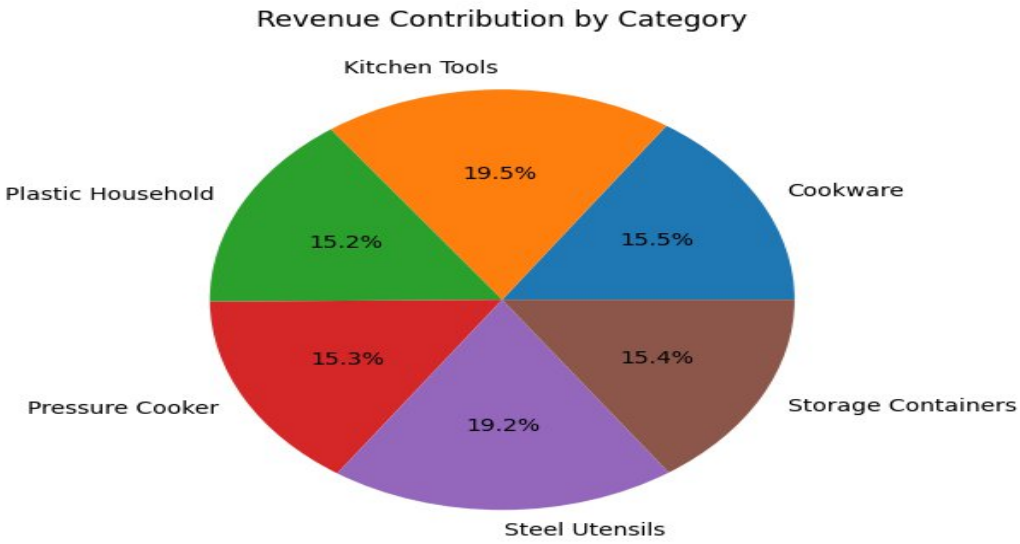
Mean = (Sum of all observations) / Number of observations

Step 3: Monthly Sales Trend Analysis

Daily sales data was aggregated monthly to identify seasonal patterns and peak sales periods.

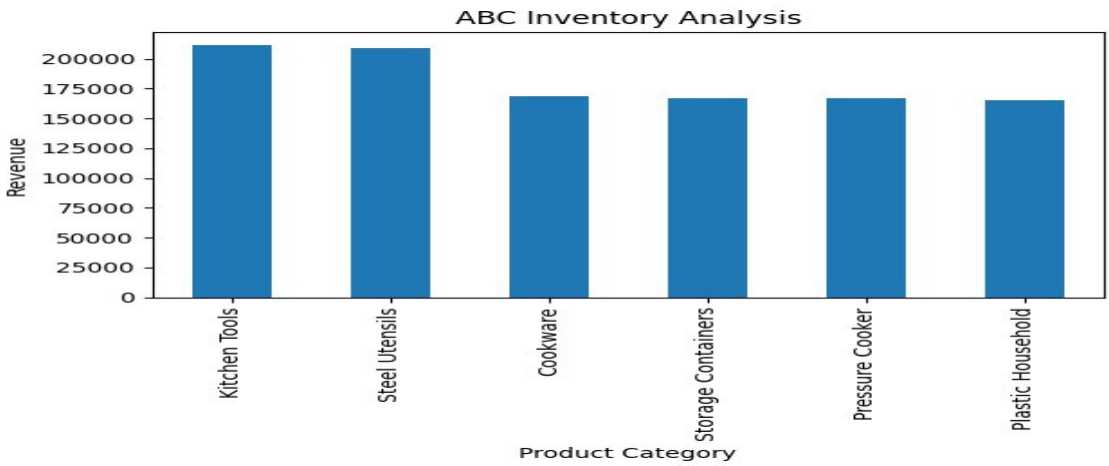
Step 4: Category Revenue Analysis

Revenue contribution of each product category was calculated to identify high-performing categories. It will help to identify the slow moving products to declutter the inventory.



Step 5: ABC Inventory Analysis

Products were categorized into A, B, and C groups based on revenue contribution to prioritize inventory management.



5. Results and Findings

The sales pattern indicated the seasonal based variation.

Sales increase during October and November due to the Diwali festival season and remain strong during December and January. The lowest sales occur during summer time June and July. ABC analysis shows that a small number of products generate the majority of revenue.

Figure 1: Monthly Sales Trend vs Daily Sales

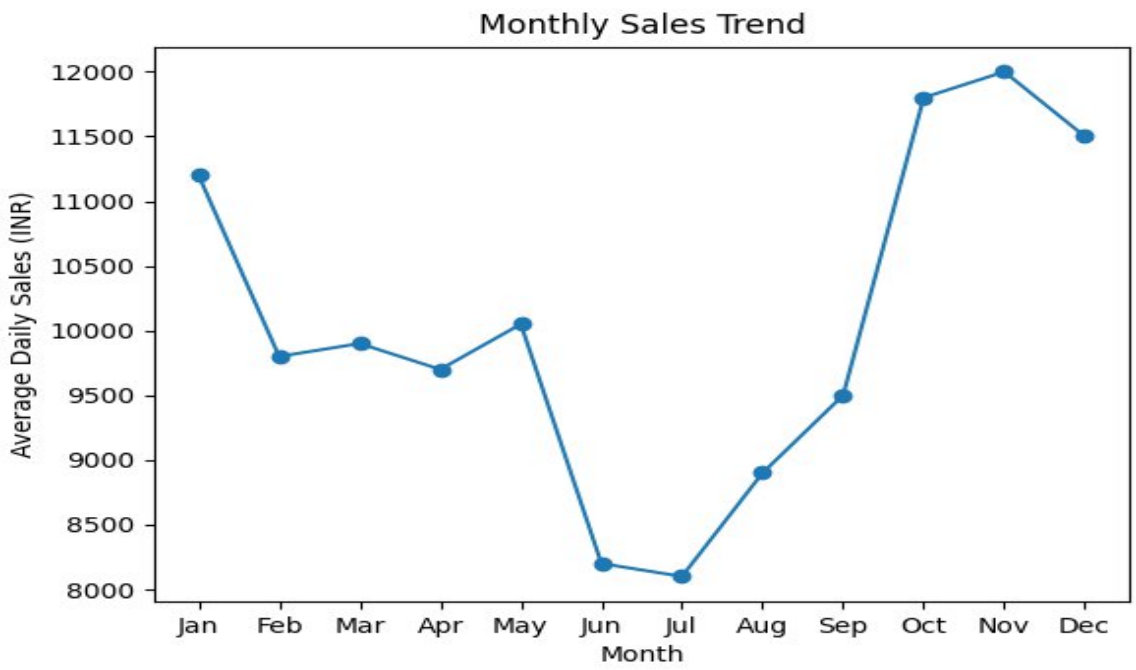
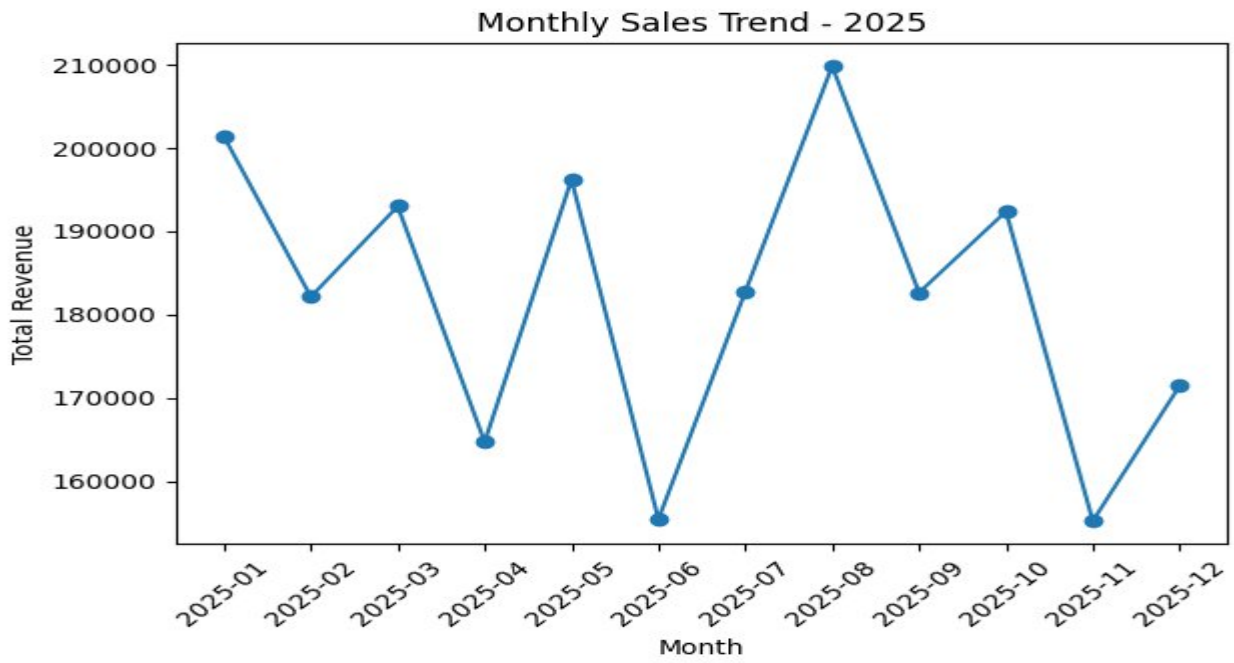


Fig: Monthly Revenue Trend



Interpretation:

The monthly sales graph illustrates the variation in revenue across different months in 2025. The sales remain relatively strong in January (₹201,373) and March (₹193,020), indicating stable demand at the beginning of the year.

A noticeable decline occurs in April (₹164,737) and June (₹155,473), which may correspond to lower consumer purchasing during summer months. Sales begin to recover from July onwards (₹182,757).

The highest revenue is recorded in August (₹209,823), indicating peak demand possibly due to festival preparation or seasonal purchasing patterns.

Another decline is observed in November (₹155,262) before recovering slightly in December (₹171,496).

From the first level analysis, we can conclude that the shop experiences moderate seasonality in sales patterns. Therefore, maintaining adequate stock levels during high-demand months such as January, May, August, and October can help the shop avoid stock shortages and improve customer satisfaction.

Product Category – Steel Utensils Descriptive Analysis

	Total Sales (₹)	Quantity Sold	Unit Price (₹)
count	185.000000	185.000000	185.000000
mean	1087.351351	2.321622	468.217391
std	742.563842	1.102173	235.456281
min	150.000000	1.000000	150.000000
25%	540.000000	1.000000	250.000000
50%	980.000000	2.000000	420.000000
75%	1560.000000	3.000000	650.000000
max	3675.000000	5.000000	1200.000000

These findings suggest that steel utensils are consistently sold products, contributing steadily to the shop's revenue. Therefore, maintaining adequate inventory levels for this category would help ensure that customer demand is met without stock shortages.

--- I will further add additional analysis and graphs for final report ---

Thank You!